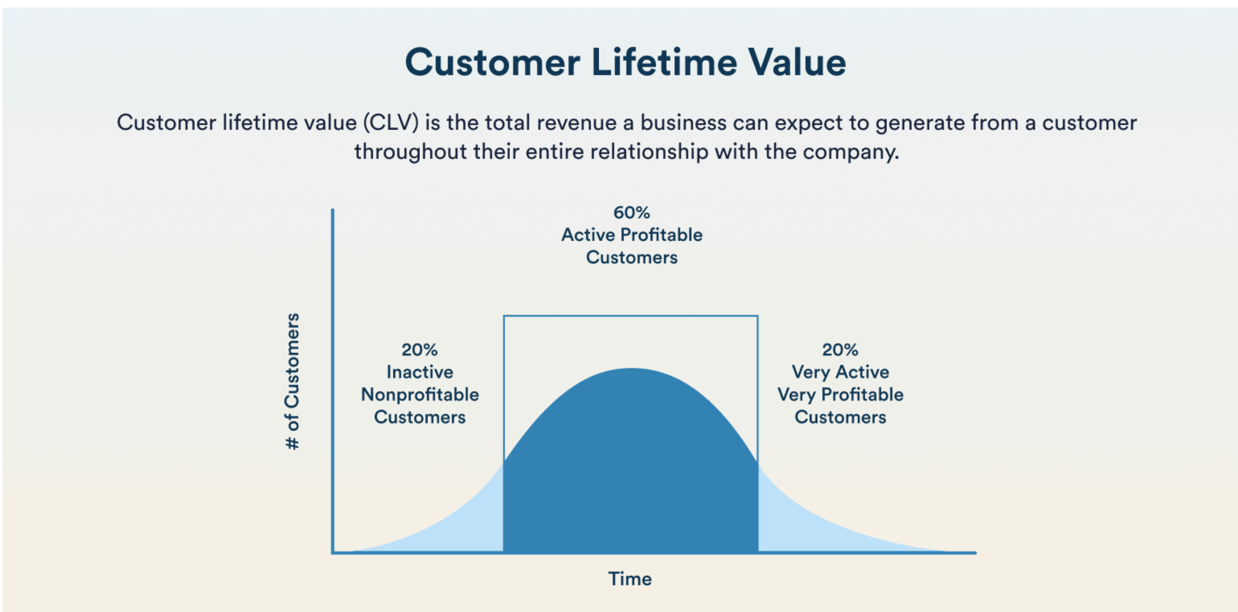
# Customer Lifetime Value Analysis

*Machine Learning Comprehensive Report*

**Dipanshu Kumar**




## EXECUTIVE SUMMARY

This project conducted a comprehensive Customer Lifetime Value (CLV) analysis on an online retail dataset spanning December 2010 to December 2011. Through RFM segmentation, historical CLV calculation, and predictive modeling, we analyzed 3,874 customers to identify high-value segments and build a forecasting model for future customer value. Key findings reveal that 27.6% of customers (Champions) contribute 72.7% of total revenue, with an overall 6-month retention rate of 69.6%. Our XGBoost prediction model achieves 51.4% accuracy on test data, enabling data-driven customer prioritization and strategic retention initiatives projected to generate £145,991 in additional revenue (2.6% ROI).

# INTRODUCTION & PROBLEM STATEMENT

## BACKGROUND

Customer Lifetime Value (CLV) represents the total revenue a business can expect from a customer throughout their relationship. Understanding CLV enables companies to optimize marketing spend, prioritize customer retention efforts, and allocate resources efficiently toward high-value customers.

## OBJECTIVES

- Segment customers using RFM (Recency, Frequency, Monetary) analysis
- Calculate historical CLV across a 6-month prediction window
- Build a predictive model to forecast future customer value
- Provide actionable business recommendations for customer management

## DATASET OVERVIEW

- Source: Online Retail Dataset
- Time Period: December 1, 2010 – December 9, 2011
- Initial Records: 541,909 transactions
- Final Records: 347,946 transactions (after cleaning)
- Unique Customers: 3,874
- Total Revenue: £5,614,837.20
- Geographic Focus: United Kingdom (495,478 transactions, 91.4%)

first 5 records:

| | InvoiceNo | StockCode | Description | Quantity | InvoiceDate | UnitPrice | CustomerID | Country |
|---|---|---|---|---|---|---|---|---|
| 0 | 536365 | 85123A | WHITE HANGING HEART T-LIGHT HOLDER | 6 | 2010-12-01 08:26:00 | 2.55 | 17850.0 | United Kingdom |
| 1 | 536365 | 71053 | WHITE METAL LANTERN | 6 | 2010-12-01 08:26:00 | 3.39 | 17850.0 | United Kingdom |
| 2 | 536365 | 84406B | CREAM CUPID HEARTS COAT HANGER | 8 | 2010-12-01 08:26:00 | 2.75 | 17850.0 | United Kingdom |
| 3 | 536365 | 84029G | KNITTED UNION FLAG HOT WATER BOTTLE | 6 | 2010-12-01 08:26:00 | 3.39 | 17850.0 | United Kingdom |
| 4 | 536365 | 84029E | RED WOOLLY HOTTIE WHITE HEART. | 6 | 2010-12-01 08:26:00 | 3.39 | 17850.0 | United Kingdom |

*Fig: First 5 records*

# METHODOLOGY

## DATA PREPROCESSING

Initial Data Quality Assessment:

- Missing CustomerID: 134,697 rows (24.8%)
- Cancelled Orders: 9,288 rows (1.7%)
- Invalid Prices: 40 rows
- Quantity/Price Outliers: 6,378 rows

```
Data quality assessment
             Missing_Count  Missing_Percentage       Data_Type
InvoiceNo                0            0.000000          object
StockCode                0            0.000000          object
Description           1454            0.268311          object
Quantity                 0            0.000000           int64
InvoiceDate              0            0.000000  datetime64[ns]
UnitPrice                0            0.000000         float64
CustomerID          135080           24.926694         float64
Country                  0            0.000000          object

additional quality checks:
Duplicate rows: 5268
Negative quantities: 10624
Zero/negative prices: 2517
Cancelled orders (C prefix): 9288
```

*Fig: Initial data quality*

Cleaning Steps:

1. Removed cancelled orders (InvoiceNo starting with 'C')
2. Dropped rows with missing CustomerID
3. Removed transactions with negative/zero quantities or prices
4. Filtered to UK market (primary revenue source)
5. Removed statistical outliers (>99th percentile for quantity and price)

Result: 35.8% data reduction, improved quality for analysis

```
Data preprocessing pipeline:
Initial shape: (541909, 9)
Removed 9,288 cancelled orders
Removed 134,697 rows with missing CustomerID
Removed 0 rows with negative/zero quantities
Removed 40 rows with negative/zero prices
Added Revenue column
Filtered to UK market: 354,321 transactions
Removed 3,288 quantity outliers (>P99)
Removed 3,090 price outliers (>P99)

Final preprocessed shape: (347946, 9)
Data quality improvement: 35.8% reduction

Preprocessed dataset summary:
Date Range: 2010-12-01 08:26:00 to 2011-12-09 12:49:00
Unique Customers: 3,874
Unique Products: 3,595
Total Revenue: £5,614,837.20
Average Order Value: £347.82
```

*Fig: Preprocessed Data*

## RFM SEGMENTATION

RFM scores each customer on three metrics (1-5 scale):

- Recency: Days since last purchase (lower = better)
- Frequency: Number of transactions (higher = better)
- Monetary: Total spending (higher = better)

Combined RFM scores segment customers into 10 distinct groups, each representing different value and retention characteristics.

```
RFM summary statistics:
            Recency    Frequency      Monetary
count   3874.000000  3874.000000   3874.000000
mean      92.262261     4.167011   1449.364276
std       99.620635     6.948056   3319.662680
min        1.000000     1.000000      2.900000
25%       18.000000     1.000000    276.682500
50%       51.000000     2.000000    606.790000
75%      143.750000     5.000000   1473.295000
max      374.000000   204.000000  71564.790000
```

*Fig: RFM summary*

```
R_Score distribution:
R_Score
1    771
2    773
3    746
4    803
5    781
Name: count, dtype: int64
```

```
F_Score distribution:
F_Score
1    1345
2     741
3     454
4     560
5     774
Name: count, dtype: int64
```

```
M_Score distribution:
M_Score
1    775
2    775
3    774
4    775
5    775
Name: count, dtype: int64
```

*Fig: R,F,M scores distribution*

## HISTORICAL CLV CALCULATION

- Training Period: Dec 2010 – May 2011 (6 months)
- Prediction Period: Jun 2011 – Dec 2011 (6 months)
- Features Engineered: Purchase rate, revenue per day, customer lifespan
- Target Variable: Actual 6-month CLV (revenue in prediction period)

## PREDICTIVE MODELING

Model Selection: **XGBoost Regressor**

- Rationale: Handles non-linear relationships, interpretable feature importance
- Features: Purchase_Frequency, Total_Revenue, Avg_Order_Value, Customer_Lifespan, Purchase_Rate, Revenue_Per_Day, Total_Quantity
- Validation: 80/20 train-test split with 5-fold cross-validation

```
CLV prediction model:
Features: ['Purchase_Frequency', 'Total_Revenue', 'Avg_Order_Value', 'Customer_Lifespan', 'Purchase_Rate', 'Revenue_Per_Day', 'T
otal_Quantity']
Training samples: 2,427
Average target value: £986.05
Training set: 1,941 samples
Test set: 486 samples
```

*Fig: Model features*

# KEY RESULTS

## EXPLORATORY DATA ANALYSIS

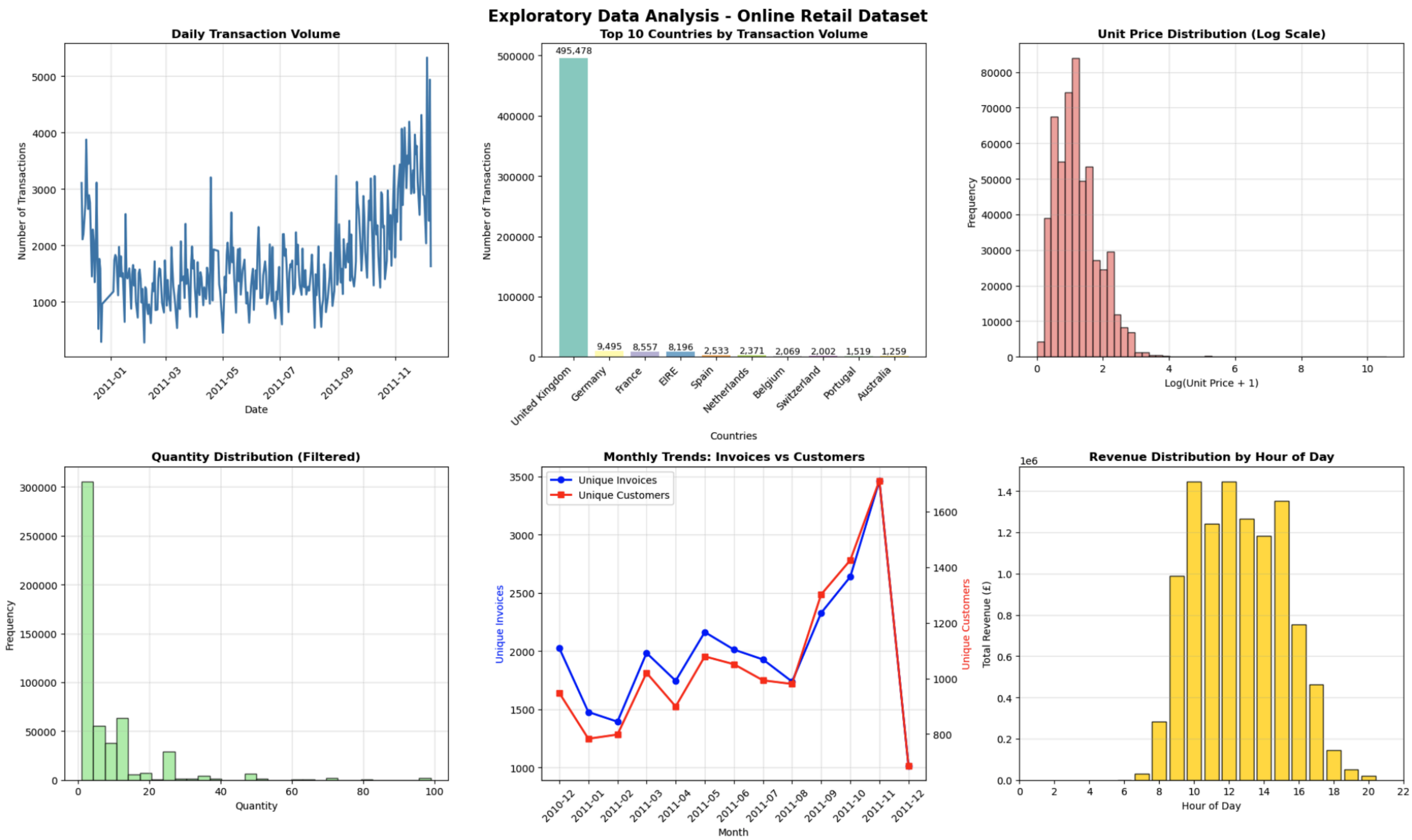



*Fig: EDA visualisations*

**Key Insights:**

- Peak transaction day: 2011-12-05 (5,331 transactions)
- Top country: United Kingdom (495,478 transactions)
- Average unit price: £4.67
- Peak business hour: 10:00 (£1,446,742.70 revenue)
- Total revenue: £10,666,684.54

# RFM SEGMENTATION RESULTS

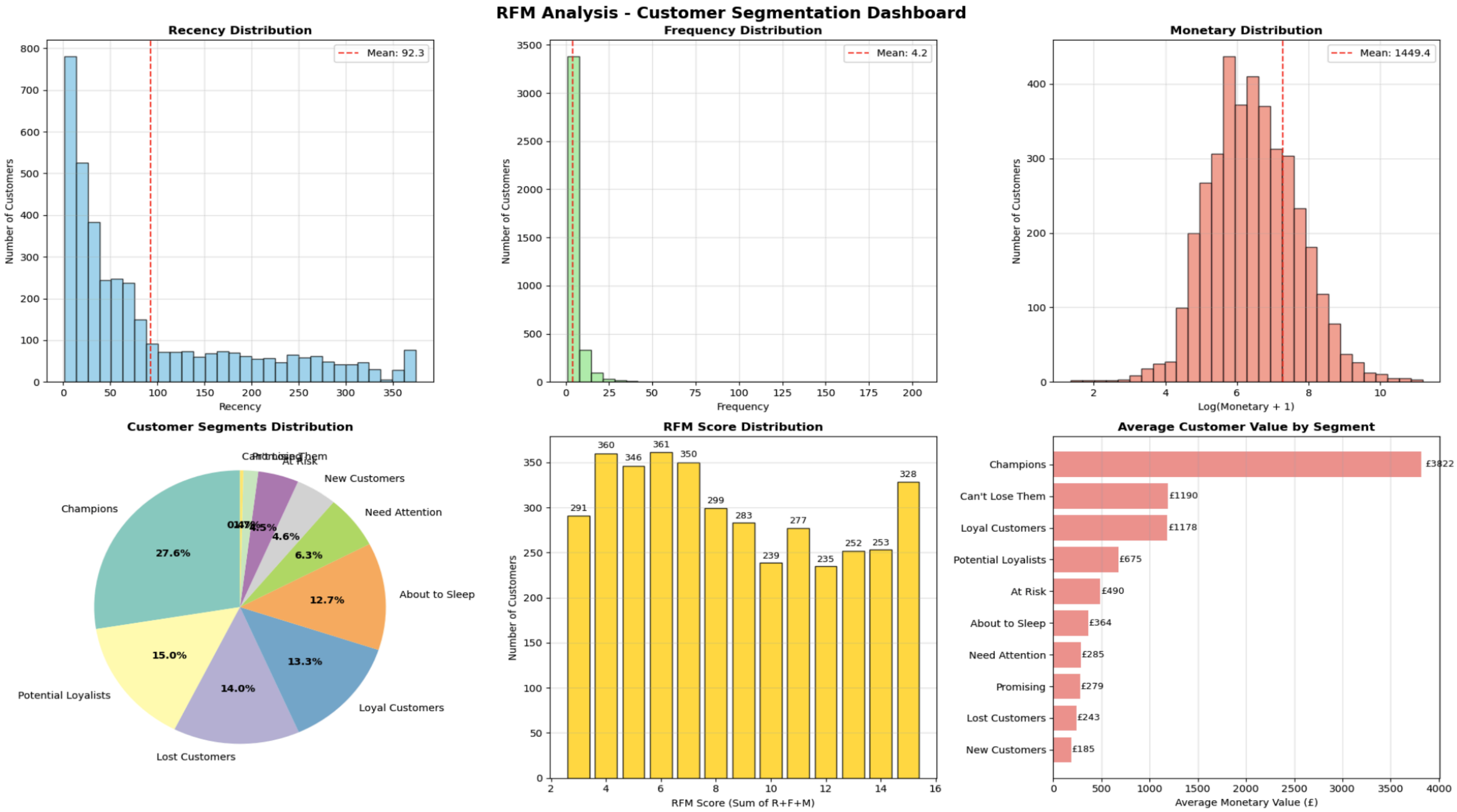



*Fig: RFM visualisations*

| Segment | Customers | % of Base | Avg Revenue | Revenue Share |
|---|---|---|---|---|
| Champions | 1,068 | 27.6% | £3,821.86 | 72.7% |
| Loyal Customers | 516 | 13.3% | £1,177.74 | 10.8% |
| Potential Loyalists | 582 | 15.0% | £674.61 | 7.0% |
| About to Sleep | 492 | 12.7% | £363.63 | 3.2% |
| Lost Customers | 543 | 14.0% | £242.50 | 2.3% |
| At Risk | 174 | 4.5% | £489.67 | 1.5% |
| Other Segments | 499 | 13% | £232–£1,190 | 1.5% |

*Table: RFM segmentation*

**Key Insights:** 27.6% of customers generate 72.7% of revenue, demonstrating extreme customer value concentration.

## CLV ANALYSIS

Historical 6-Month CLV:

- Total Customers Analyzed: 2,427
- Customers with Positive CLV: 1,690 (69.6% retention)
- Average 6-Month CLV: £986.05
- Median 6-Month CLV: £370.45
- Top 20% Customers Contribution: 72.1% of total CLV

Correlation Analysis:

- Frequency vs CLV: 0.607 (strong positive)
- Customer Lifespan vs CLV: 0.350 (moderate positive)

Risk Assessment:

- At-Risk Customers: 1,209 (31.2%)
- Potential Revenue at Risk: £395,787 (7.0% of total)

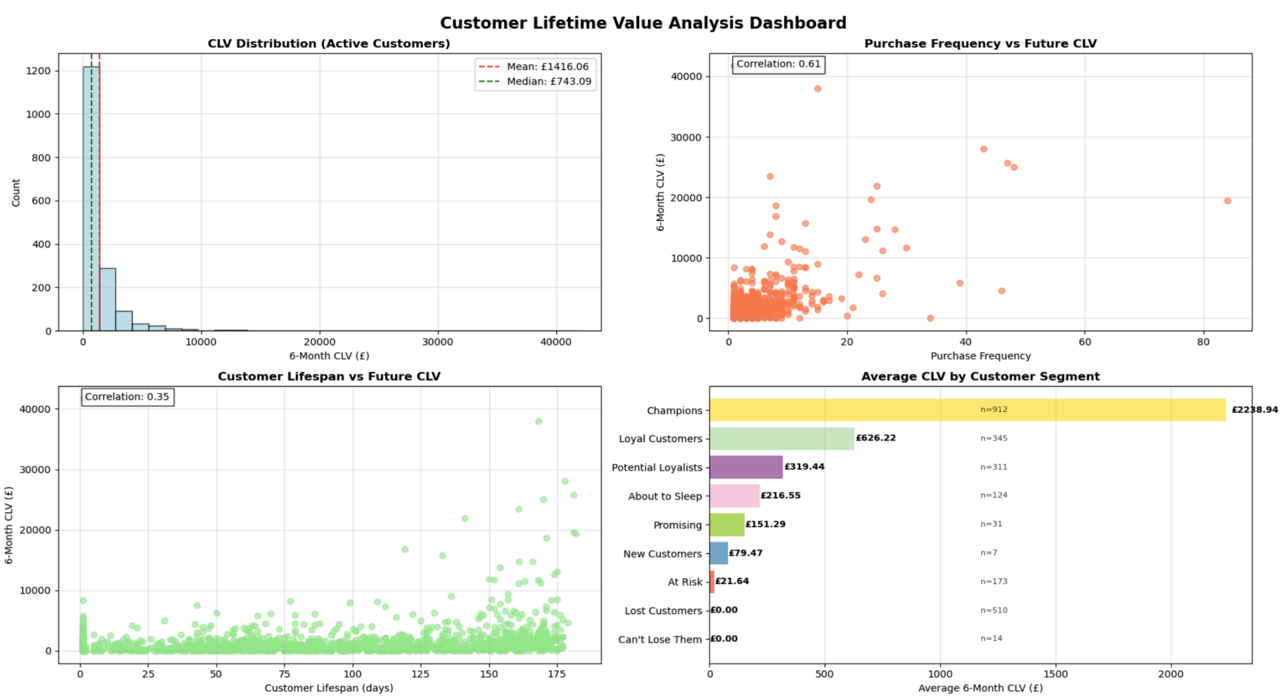



*Fig: CLV analysis*

# PREDICTION MODEL PERFORMANCE

Model Metrics:

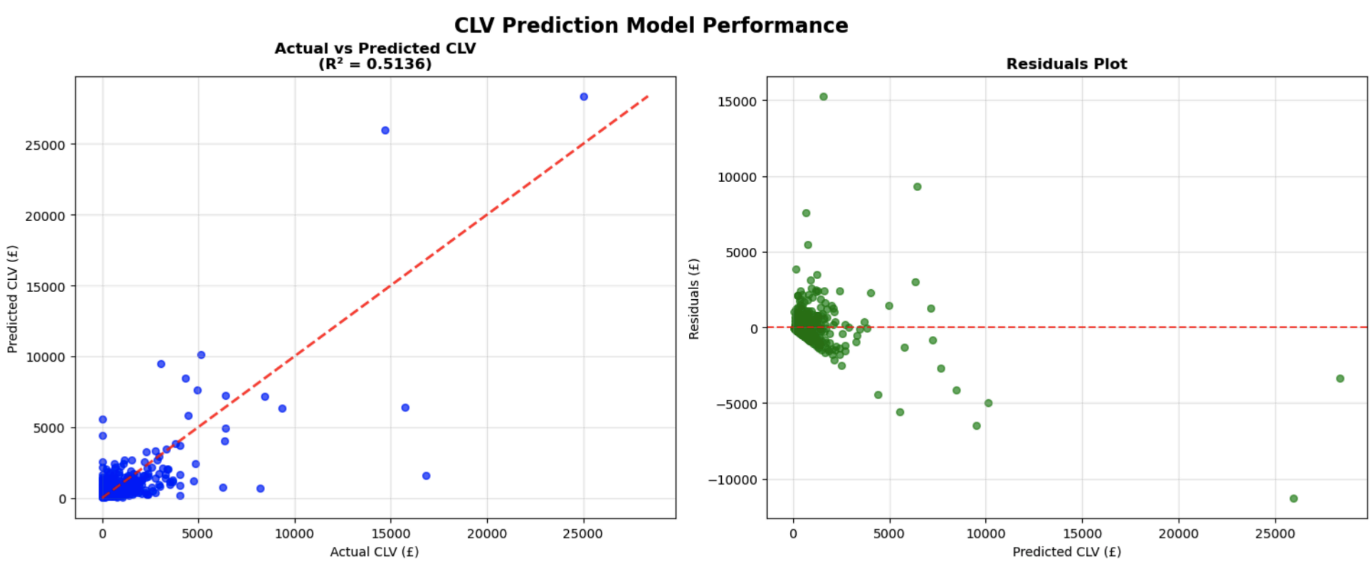
- Training $R^2$: 0.9604 (excellent fit on training data)
- Test $R^2$: 0.5136 (moderate predictive power)
- Test RMSE: £1,421.74
- Mean Absolute Error: £688.22
- Cross-Validation $R^2$ Mean: 0.5295 (±0.2706)

Prediction Accuracy:

- Within ±£100: 14.4%
- Within ±£500: 62.3%
- Median Error: £355.38

Feature Importance:

1. Total_Revenue: 62.28%
2. Revenue_Per_Day: 20.06%
3. Customer_Lifespan: 4.48%
4. Total_Quantity: 4.23%
5. Purchase_Frequency: 3.28%



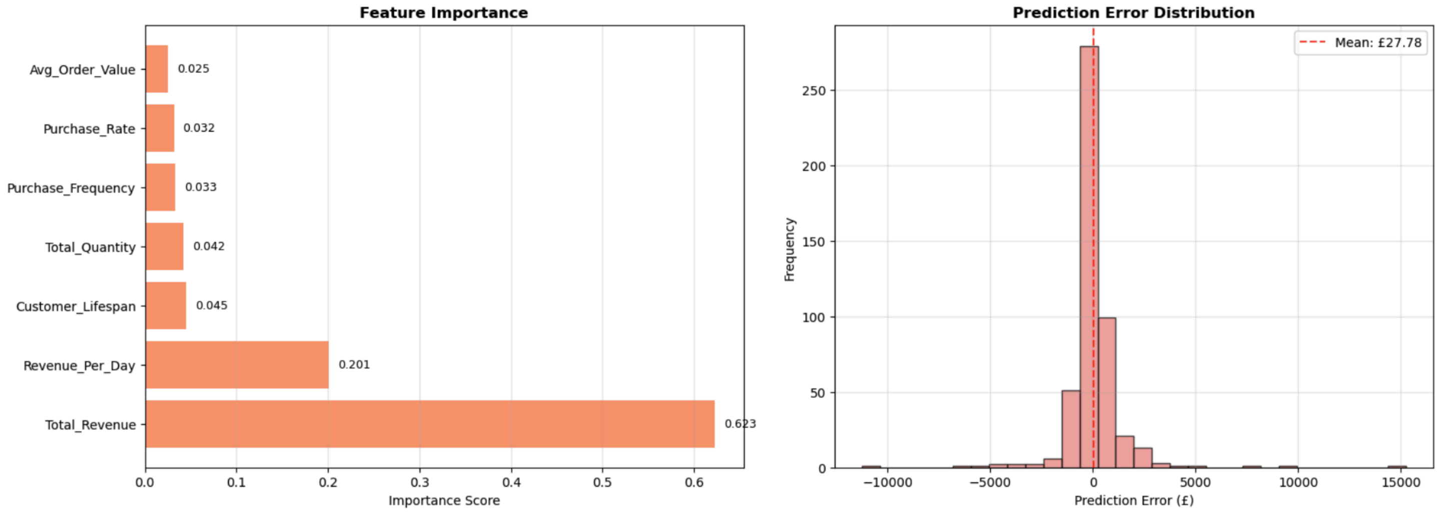



*Fig: Prediction model*

**Interpretation**: Historical revenue is the strongest CLV predictor (62%), indicating past spending patterns are highly informative for future value.

# CUSTOMER CLUSTERING ANALYSIS

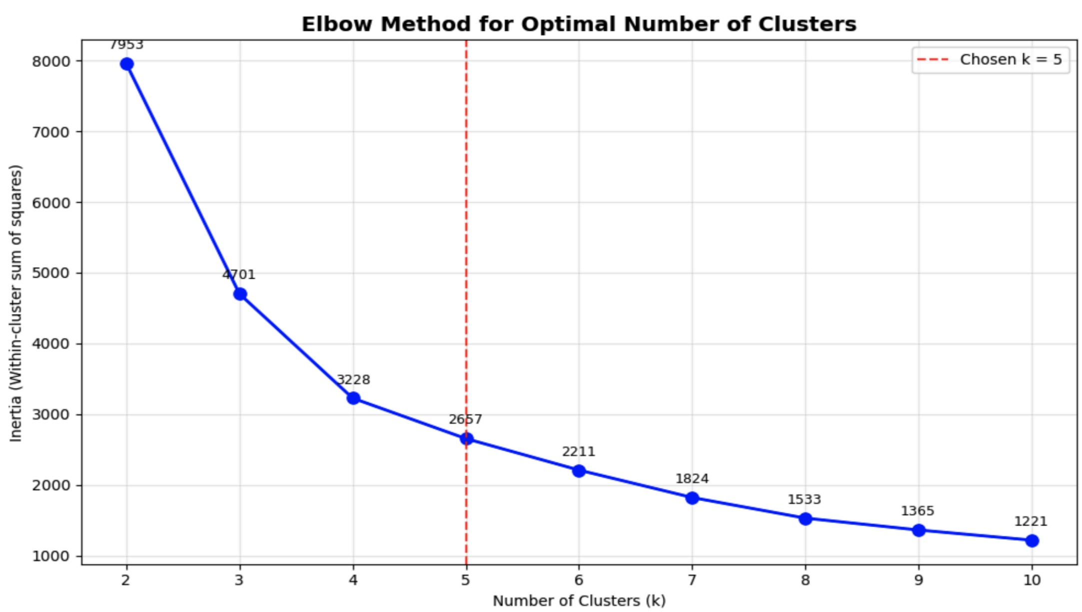



*Fig: Number of clusters*

K-means clustering (k=5) identified distinct customer behavioral groups:

| Cluster | Size | Recence | Frequency | Avg Value | Profile |
|---|---|---|---|---|---|
| Recent Customers | 2,537 | 45 days | 3.15x | £961 | Core regular customer base |
| High-Value Active | 31 | 6 days | 38x | £25,399 | VIP segment, frequent buyers |
| At-Risk Customers | 957 | 247 days | 1.55x | £401 | Churned, inactive |
| Emerging High-Value | 345 | 17 days | 14.4x | £5,293 | Growth potential |
| Ultra High-Value | 4 | 2 days | 128x | £44,736 | Extreme whales |

*Table: Customer clustering*

# BUSINESS INSIGHTS & RECOMMENDATIONS

## STRATEGIC INITIATIVES

1. Champion & Loyal Customer Retention (1,584 customers)

- Implement VIP loyalty program with exclusive benefits
- Personalized product recommendations based on purchase history
- Early access to new products and sales
- Dedicated customer service tier
- Expected Impact: Reduce churn by 5%, retain £86,333 in CLV

2. Customer Reactivation Campaign (666 at-risk customers)

- Targeted win-back email campaigns with personalized offers
- Segment messaging by inactivity level
- Special one-time discounts (10-15% off)
- Survey to understand churn reasons
- Expected Impact: 10% reactivation rate = £26,411 new CLV

3. Growth Development Program (823 potential loyalists)

- Cross-sell and upsell campaigns tailored to purchase history
- Bundle offers to increase average order value
- Loyalty point programs incentivizing repeat purchases
- Frequency-based escalating discounts
- Expected Impact: 15% increase in average spend = £33,247 growth

4. Predictive CLV Optimization

- Use model predictions to set dynamic customer acquisition costs (CAC) limits
- Allocate support resources to high-predicted-value customers
- Implement predictive churn scoring for early intervention
- Price optimization based on CLV tier

## PROJECTED FINANCIAL IMPACT

| Initiative | Revenue Impact | Rationale |
| --- | --- | --- |
| Retention Improvement | £86,333 | 5% improvement on 1,584 customers |
| Reactivation | £26,411 | 10% success rate on 666 targets |
| Upselling | £33,247 | 15% AOV increase on 823 customers |
| Total Projected | £145,991 | 2.6% ROI vs current revenue |

*Table: Financial impact*

# MODEL LIMITATIONS & FUTURE ENHANCEMENTS

## LIMITATIONS

1) **$R^2$ of 0.514:** Model explains 51% of CLV variance; 49% driven by external factors
2) **Geographic Constraint:** UK-only analysis; generalization to other markets uncertain
3) **Temporal Scope:** 12-month dataset; longer-term patterns may differ
4) **Feature Scope:** Additional features (seasonality, product category, customer demographics) could improve accuracy

## FUTURE ENHANCEMENTS

1. Incorporate seasonality and product category indicators
2. Test advanced models (LightGBM, neural networks) for improved $R^2$
3. Expand to multi-country analysis
4. Implement customer lifetime value prediction at transaction level
5. Develop real-time churn prediction system
6. Integrate customer satisfaction scores (NPS, reviews)

# CONCLUSION

This CLV analysis successfully identified that 27.6% of customers (Champions) generate 72.7% of revenue, providing clear prioritization for resource allocation. The 69.6% retention rate demonstrates strong customer loyalty, though 31.2% at-risk segment requires proactive reactivation efforts. Our predictive model, while achieving moderate accuracy ($R^2$ = 0.514), provides directional guidance suitable for strategic planning and customer segmentation.

**Key Takeaway:** Implementing the recommended initiatives; focused retention, targeted reactivation, and strategic development programs can deliver £145,991 in incremental revenue (2.6% ROI) through data-driven customer management.

Recommended Next Steps:

1. Implement Champion VIP program immediately (highest ROI)
2. Launch reactivation campaign for at-risk segment
3. A/B test upsell messaging on Potential Loyalists
4. Monitor model predictions monthly and retrain quarterly
5. Integrate predictive CLV into CRM system for operational use

# TECHNICAL SUMMARY

Tools & Technologies: Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn

Analysis Techniques: RFM Segmentation, K-means Clustering, Feature Engineering, Cross-validation, Regression Modeling

Deliverables:

- Segmented customer dataset (3,874 customers × 15 features)
- Trained predictive model with performance metrics
- Business recommendations with ROI projections
- Production-ready Python notebook for model retraining

---

*Report Generated: November 2025*

*Dataset Period: December 2010 – December 2011*

*Analysis Scope: 3,874 customers, 347,946 transactions, £5.6M revenue*